THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET B

Chapter 5: **Life Processes** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The tiny pores on leaves through which gases are exchanged are called: 1

(a) veins **(b)** root hairs **(c)** stomata **(d)** lenticels

2. Which of these is an autotroph? 1

(a) green plant **(b)** fungus **(c)** lion **(d)** cuscuta

3. The first step of glucose breakdown, forming pyruvate, takes place in: 1

(a) mitochondria **(b)** cytoplasm **(c)** nucleus **(d)** chloroplast

4. A student states: “Platelets help in blood clotting.” The correct response is: 1

(a) Correct **(b)** Only in plants

(c) Incorrect **(d)** Cannot be determined

5. For the function of the lymph, Student A says it carries digested fat; Student B says it pumps blood. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. Haemoglobin is present in: 1

(a) red blood cells **(b)** white blood cells

(c) plasma **(d)** platelets

7. Which organism uses parasitic nutrition? 1

(a) cuscuta **(b)** mushroom **(c)** green plant **(d)** amoeba

8. The energy currency of the cell is: 1

(a) glucose **(b)** ATP **(c)** ADP **(d)** NADP

9. The opening of the stomach into the small intestine is regulated by: 1

(a) epiglottis **(b)** sphincter muscle **(c)** diaphragm **(d)** villi

10. Urine is carried from the kidneys to the urinary bladder by the: 1

(a) renal artery **(b)** urethra **(c)** collecting duct **(d)** ureters

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. How do amoeba obtain their food? 2

12. Name the two components of the transport system in plants. 2

13. Why do aquatic animals breathe faster than terrestrial animals? 2

14. A student says that the right side of the heart carries oxygenated blood. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. What is the role of acid in the stomach and of the liver in digestion? 3

16. Explain the mechanism of opening and closing of stomata. 3

17. How are fats digested in our bodies? Where does this process take place? 3

OR

What is translocation? Why is it essential for plants? How does it occur? 3

18. What is dialysis? Why is it needed? 3

OR

How do plants get rid of their excretory products? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Describe the human excretory system with a labelled diagram. 5
(ii) Explain the process of urine formation in a nephron.
(iii) How is the amount of urine regulated?

OR

(i) Explain the different ways glucose is broken down in organisms with a flow chart. 5
(ii) Where do these processes occur?

20. **Case study:** Lymph is a fluid that leaks from capillaries into intercellular spaces. 5
(i) How does lymph differ from blood?
(ii) Name two functions of lymph.
(iii) What are the components of blood?
(iv) What is the function of platelets?

OR

(i) Describe the alimentary canal of man (main parts). 5
(ii) What is peristalsis?
(iii) Function of the large intestine?

END OF QUESTION PAPER — Total: 40 Marks

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